

Sidarth Kulkarni

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science and Financial Technology, Deans List

Aug. 2021 – Dec. 2025

Relevant Coursework: Data Structures and Algorithms, Object Oriented Design, Computer Systems (C), Machine Learning, Software Engineering (Microservice Architecture), Database Design, Law and Ethics of Technology

EXPERIENCE

Software Engineer

Aug. 2024 – Present

Docipher

Remote

- Designed and deployed AI-powered medical Q&A system adopted by **15+ patients** and **10+ doctors**, reducing medical query resolution time.
- Fine-tuned Google Gemini with semantic guardrails, reaching **91% accuracy** on clinical benchmark tasks.
- Built scalable backend (Python, SQL) supporting <200ms response latency for 20+ monthly queries. Dockerized on GCS.

Software Engineer Co-op

Jan. 2025 – June 2025

MORSE Corp

Cambridge, MA

- Developed runtime validation system for ML algorithms, boosting data quality compliance from **23%** to **95%**.
- Optimized YOLOv8 object detection pipeline in **PyTorch**, reducing GPU memory usage by **76%**.
- Led **3-person agile team** to deliver AI-powered training data assessment tool in **66%** of allocated timeline.
- Built analytics pipeline in **Python & Polars** to design and visualize new object detection metrics and quantify product efficacy.

Quantitative Analytics Intern

May 2024 – Aug. 2024

PNC Bank

New York City, NY

- Developed Gradient Boosting Classifier with **0.81 AUC** to predict delinquency patterns, presenting insights to executives.
- Engineered automated feature selection pipeline with **PyCaret**, improving AUC by **2%**. Worked on Linux based cluster to manage and queue training and inference jobs.
- Built SQL + Python analytics pipeline for real-time C&IB investment tracking, reducing decision-making time by **63%**.

Technical Project Management Intern

June 2023 – May 2024

SiPhox Health

Burlington, MA

- Developed full-stack web application (**React**, **ExpressJS**, CRUD) to track and manage orders.
- Prototyped a full-stack application in **TypeScript** using Typeform and Zapier APIs, which increased monthly revenue by \$4K with a 4% conversion rate.
- Filed provisional patent for auxiliary blood collection device with **100% success rate**; oversaw design (**AutoCAD**) and manufacturing.

PROJECTS

RAG Benchmarking Tool | *Python, ClickHouse, PostgreSQL, Docker, Langchain*

June 2025 – July 2025

- Developed benchmarking system for evaluating vector database performance in **Retrieval-Augmented Generation** tasks using Claude over 3.1k test queries.
- Implemented containerized microservices with Docker, enabling asynchronous workers on NetApp cluster, reducing decision making time by **40%** compared to simultaneous queries.

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, VBA, C, C++, Bash

Tools: React, MongoDB, PostgreSQL, Docker, Git, LLMs, CI/CD, Linux, Figma, GCS, AWS, PowerBI

INTERESTS

Electric Guitar, Tennis, 3D Modeling, UX Design, Running, Jazz Piano